Why Craters Are Round, Size of Life, The Real Length of Day – TYTYK

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis: This episode utilizes astrophysics, celestial mechanics, and biological scaling laws to explain fundamental physical principles that govern the universe. It corrects long-standing misconceptions regarding the origin of lunar craters (attributed to high-speed impacts rather than volcanism), clarifies the discrepancy between Earth's rotational periods (sidereal vs. solar days) and the necessity of atomic timekeeping (leap seconds), and establishes the square-cube law as a constraint on biological existence, defining why size dictates physical capabilities from surface tension in insects to structural integrity in elephants.

2. Key Arguments & Data:

3. **Lunar Crater Formation:** All lunar craters are perfect circles because meteoroids impact at such high speeds that kinetic energy exceeds the object's binding energy. Upon impact, the energy is converted instantly into vaporization (90% of the asteroid), creating a circular cavity regardless of the angle of incidence.
4. **The "Magic Speed":** High-speed collisions occur when an object's velocity causes it to explode upon impact. While the transcript humorously references ~89 mph in relation to flux capacitors, the physical claim is that at specific high speeds relative to the target, the energy release guarantees a circular shape even at oblique angles (compared to Earth, where lower speeds allow surface erosion).
5. **Meteor Crater Arizona:** Originally misidentified as volcanic due to its perfect circle and flatness ("Seas of Tranquility"), this site was later identified as an impact crater (Barringer). Computer simulations from the 1970s proved that if found, the original asteroid would have been vaporized rather than buried.
6. **Sidereal vs. Solar Day:** Earth completes one rotation relative to fixed stars in exactly 23 hours, 56 minutes, and 4 seconds (sidereal day). The solar day is 24 hours because Earth must rotate an additional ~1 degree to account for its orbit around the Sun during that rotation (~360° / 365 days).

7. **Timekeeping and Rotation Stability:** Time is now defined by atomic transitions (Cesium-133 electron oscillations), not Earth's rotation. Due to tidal friction from the Moon, Earth's rotation slows, requiring leap seconds to remain synchronized with atomic time (~24 added since the early 70s).
8. **Orbital Variance:** The solar day length varies throughout the year as Earth orbits closer to or farther from the Sun due to elliptical orbit mechanics (speed changes per Kepler's laws), necessitating a map correction (analemma) on sundials.
9. **Square-Cube Law in Biology:** Strength scales with cross-sectional area (r^2), while mass scales with volume (r^3). As size increases, the surface must support disproportionately more weight. This explains why elephants have thick legs but insects can maintain thin limbs to support their own weight relative to their strength.
10. **Surface Tension:** In macroscopic environments (human scale), gravity dominates over surface tension. In microscopic environments (insect scale), surface tension dominates, allowing organisms to walk on water or drink from a bead of fluid without it spreading.

11. **Scaling Limits for Life:** Biological signaling is limited by the speed of light; neurons cannot transmit signals fast enough across a galactic-sized organism, making macroscopic bodies functionally impossible. Conversely, sub-molecular sizes are impossible due to quantum chemistry and molecular binding limits.

12. Actionable Takeaways:

13. **Time Awareness:** Understand that Earth's rotation is a variable quantity requiring correction (leap seconds); rely on atomic definitions for precision rather than solar rotation for high-stakes timekeeping (e.g., banking or navigation).
14. **Biological Contextualization:** Recognize the physical constraints of size when analyzing animal physiology; appreciate why large land animals require robust skeletal structures while microscopic organisms do not face weight-borne stress.
15. **Force Dominance Awareness:** When dealing with fluid dynamics or material science, recognize which force (gravity vs. surface tension)

dictates behavior based on the scale of the object being analyzed.

Full Transcript

Thanks for showing up for these. You know how I come up with them? There are things that people know about, but they don't know as much as they think they should know or might want to know after they learned all they could know. That was very Dr. Sussi about you. Welcome to Star Talk, your place in the universe where science and pop culture collide. Star Talk begins... right now. Something very simple. The shape of craters. Shape of craters. Yeah. Yeah. A picture of the moon. Any photograph of the moon. Right. Shape of the craters. Well, they're all circle concave little circles. They're perfect circles. Yeah. Every single one of them. The big ones, the little ones, the medium ones. They are perfect circles. Okay. So let me guess. Let me guess.

Aliens, just like on Earth, same way they make the crop circles. They make moon circles. That's how they make the moon circles. Moon circles. So, yeah, here's the thing. It's interesting that as humans, if there's something scientists don't understand, people rapidly just go to God or aliens. Well, that's why they got the explanation for what. All right. It turned out it was a mystery. why all craters are perfect circles for the longest time. It was a mystery until about a hundred years ago. Okay. And let me tell you why it was a mystery. All right. So your first thought is, well, the moon doesn't have an atmosphere. Maybe it's getting slammed by meteors. Right. But is every meteor coming straight down at a 90 degree angle so that the explode ready hits it makes a prayer is that surely some of them are at an angle you'd expect some craters to be elongated right and you'd have all sort of shapes from circle to very uh in like very flattened circles you'd expect that some of them would be divots like like when you swing a golf you know it'd just be like a Oh, okay. I mean, it came in on the angle on the angle from and with the with the because you'd have a the slope would be different on one side than the other than on the other because it came slid in is slid in. So you'd expect that and it is nowhere to be found on the moon. So this energized the geologists because we know that the dark areas of the moon is where lava has once flowed. Okay. Cool. I think they call it basalt. It's dark and it's lava fields. All right.

And by the way, the Apollo astronauts aimed for those places because they're flat. If you're going to go a quarter million miles and land sideways and fall over, that's it. So they found the flattest areas. And those are called seas. Before people knew that the moon didn't have weather like we have or oceans, these large flat areas were called sea of tranquility. Okay, that's where it gets its name. Nice. And all the C's were named after psychological states or physiological states. So there's a sea of fecundity. There's a sea... Oh, all right. Yeah, all right. So anyway, so they're flat because lava in liquid state flattens out, right? There you go. Okay. All right. So wherever there was a crater, it was round. So the geologists said it's probably not these asteroids because they'd have to all come in perpendicular.

they're probably all volcanic craters because volcanic craters you punch up it makes a hole the hole is a circle right okay you know you can you can pull that off so that's that's how it stood for the longest while until people saw craters on top of places where it didn't look like lava had flowed So how are you going to have a crater and no lava signature anywhere near it? What's up with that? And so this was a conundrum. I have a book from 1890 and it says, believed by many to be volcanic in origin, perhaps they're actually of asteroid impacts we are not sure. By the way, it was that way for most of the 20th century.

until computers came along and you could simulate impacts. Nice. You want to simulate it. Okay, so here you go. You ready? If you have an impactor that comes in straight into a surface, it makes a circular crater. If you send it at an angle, it will make an elongated crater. Okay? This will happen. So they're saying, no, wait a minute, let's send it in faster. Turns out, There is a magic speed with which you can send in an asteroid. And on collision, it will make a perfect circle no matter the angle it hits. No matter the angle it hits. Correct. And that's speed. OK? This is what you're... OK? You ready? Let me guess. Let me guess. 88 miles per hour. Precisely. The same speed that it takes to propel you into the...

Because the flux capacitors 89 miles per hour. All right. So here's the. So here's what's going on. Let's let's take a rock and you can ask why does the rock sort of hang together? Right? Why isn't it just decomposed into sand? Well, because there are molecules that are attached. All right. It's

molecularly attached. within itself. These are electromagnetic forces. We know what they are. You can add up how much energy is contained in all of these molecular bonds. And you can write down that number. And you say, this rock is held together by this much energy. You can do that. Some rocks are held together with not much energy. Like if you take a sort of a semi, a kind of solid thing at beach. Like a snowball. Or we'll get to that in a minute, right? Okay. Like a sand ball, right? Right. That if you just sort of punch it, the whole thing just crumbles back into sand. But it holds its shape. They held together. Just for a little while, not much energy is holding that together. Snowball, another example.

not much energy is holding unless you like really pack it in and what you're doing is you're melting the snow and have it refreeze and that holds it up better. We had a whole explainer on the freezing and melting device. So what you're really doing is saying to the other person you're having a snowball fight with, I hate you. I hate you even more. Even more. Because this is going to be a nice little ice ball hitting the face with. So that helps NFS held together only loosely. You can't even throw it without the thing flying apart. OK. So you can write down how much energy that is, calculate it and write it down. Now move the object through space or through the air. There's a kinetic energy it has. And this is the energy of motion. It's called kinetic energy. There's a formula.

It's one half times the mass of the object times the velocity squared. Okay? All right. Okay. Now, the moment the kinetic energy that the object has exceeds the binding energy of the molecules. Right. You have to ask it, well, if the thing was going at that speed and then after it hits, it's going at zero speed, then what happened to all that kinetic energy? It got pumped back into the object. Right. But that's more energy than what is holding the object together in the first place. Right. So if you have more kinetic energy than the binding energy of the thing that has the kinetic energy, it's going to explode. On impact, it explodes precisely. Wow. So we call these high-speed collisions.

And it's not just, oh, because it's going fast. No, it is higher speed than the energy that's holding it together. So on impact, it's an explosion. And explosions happen in all directions. Now, is that because the moon has no

atmosphere? Because that doesn't happen on Earth. So, so, so let's get to that in a minute. OK. Let me get to that. Just imagine there's the air is not a thing. Just imagine there's not a thing to talk about the energy. OK. Okay. All right. So, so there you have it. That's the entire reason. So in fact, all the craters on the moon, damn near all the craters on them. There might be one or two that revolve. All the rest are asteroids, meteors coming in at whatever angle they choose. Doesn't make a difference. That doesn't make a difference. They all explode on impact. They explode on impact because they are coming in at what we call a high speed collision. Now.

Now, let's look at snowballs. You should try this, okay? When you throw a snowball at a wall, the winter's coming up, okay? We're recording this in November. So you throw it at a wall. When it hits the wall, it doesn't stay as a snowball. No, it explodes. It basically pops. It goes everywhere. It goes everywhere and try this. Except for one little white dot that's stuck to the wall. Got one little white dot that's stuck to the wall. So you've done this before. Yeah, I guess. Just to all our Hawaii friends and other friends who don't have snowball fights. So try throwing it at different angles when winter comes. You will see that every time it explodes on impact and you see the scatter all around in equal directions. That's because even at your speed, okay, You're not going hypersonic speeds. It's just the speed that you throw it. That's more kinetic energy than is the binding energy of the snowball. Right. So that's why a high speed collision, you have to ask what's binding the object and how much energy does it have of motion. And it's the relationship of those two numbers that'll tell you if it's a high speed collision. So a snowball on someone's chest exploding is a high speed collision. Nice.

That's super cool. Let's just hope though that you never throw a snowball hard enough to leave a crater in somebody. That would be more. Energy then is necessary to explode the snowball at the other side. But correct. Yeah, you don't want to leave craters. All right. So let's get back to the atmosphere. Very important and interesting question. Okay. So now an asteroid comes and it sees the atmosphere. Right. Well, the atmosphere is going to slow it down a bit, isn't it? Okay. It's going to slow it down. If it slows it down enough so that it's kinetic energy is less than the energy that binds it together, It will survive on impact. Right. We do find meteorites, we

had the I.T.E. after it's on the ground, meteor coming through the air, meteorite, we find there are meteorites that make it to the ground. Okay? We have them. They're there. They slow down enough. Put the brakes on them. Put the brakes on them and they did not completely explode. Now, in the explosion, there could be some pieces that remain intact.

just in the explosion. So Meteor Crater Arizona, which used to be called Barringer Crater, and Geologists said, this has got to be volcanic because it's a perfect circle, but there's no volcanic activity in Arizona. There's no volcano for miles. So that was a conundrum. When we figured it out, it got renamed Meteor Crater. And those computer simulations were abundant in the early 70s when we were able to program computers in the science laboratories. that asteroid, they were saying, well, where is the asteroid? If an asteroid did it, let's dig and find the big asteroid, all right? And so they dug and they didn't find anything because they didn't know that the sucker blew up, okay? And so a person bought that land, a mining company bought that land on the expectation that they could mine the metals of the asteroid that must be buried deep in it and it was nowhere to, 90% of it vaporized.

on impact. There are pieces that are scattered around, and those pieces are now in collections. We have some at the museum. But most of it is gone for that reason. And that's why creators are around. There you go. OK. You know, we keep doing this. We'll run out of stuff to explain. That's like reaching the end of the internet, which I don't think it's going to happen. Well, the end of Netflix movies, right? Right. Here's yet another fact from Lost in Space, the latest collaboration between StarTalk and National Geographic. Saturn is the only planet in the solar system whose average density is less than that of water, which means if you found a bathtub big enough, Saturn would float. If you liked that fact, you can find 4,999 more in Lost in space. 5,000 facts to help navigate the universe. Lost in space is now available for pre-order, wherever books are sold. How long does Earth take to rotate on its axis? From what I've been told, 24 hours. 24 hours. Okay. In fact, we kind of defined 24 hours to mean that. Right. So that's how we divide up time. Okay.

So as you might have guessed, it doesn't take 24 hours. It actually takes 23 hours, 56 minutes and four seconds. OK. Really? Yes. Now. Wait a minute.

Hold on. So let me just be clear about what I mean. Twenty three hours. Get rid of the entire. Just get rid of the entire solar system. OK. Watch Earth rotate. OK. And stand there and time it. It will, that spot that's in front of you will come back around in 23 hours, 56 minutes and four seconds. One full revolution. Okay. Okay. That's called a sidereal day. I like this. I mean star, star day. So you're imagining with respect to the universe. Okay. Nice. However, we don't base our lives on when stars return to the same spot in the sky.

in the night sky, we'd base our days on when the sun, which also happens to be a star, returns to its spot on the sky. Okay? It turns out that takes longer than 23 hours and 56. Why? I'll tell you why. Because in the time it took Earth to rotate 23 hours and 56 minutes, it actually moved almost one degree in its orbit around the sun. So it rotates back to where it's previous light, but it has to turn a little bit extra to put the sun back in the same spot on the sky. Right. And that's a little extra. That's fantastic. Oh my God, that's just, I love that. I've only just begun. So that little extra four minutes. So I said we move a degree in our orbit each day. You know, I didn't just pull that out. How many days are there in a year?

Uh, 365. Okay, and how many degrees in a circle? 360. Yeah, so it's about a degree a day. It's about a degree a day. Yeah, to technically, it's a one degree and five, three hundred and sixty-fifths. Oh, wow. Of a degree. But it's basically a degree a day. And in that degree, you have to turn that little extra. So just imagine you keep having to turn your head just a little bit more. Just a little bit more. Okay. because you moved just a little bit, so now you gotta look back just a little bit. That's cool. Well, let's keep going, okay? Earth's orbit around the sun is not a perfect circle. Oh, by the way, so the first day is a sidereal day. The second day is a solar day. Gotcha. Okay? So for obvious reasons, and they're not the same. All right, now, Earth's orbit around the sun is not a perfect circle. Right? Which means sometimes we're farther away, sometimes we're closer. When we are closer, we are moving faster in our orbit than when we are farther away. Okay. Okay? That's how gravity works. So it turns out that extra four minutes is not the same if we're farther from the Sun than if we're closer to the Sun. Because we move more than our allocated fraction of a circle when we're close to the Sun and less when we're farther away from the Sun.

That's it. Okay. So the length of the solar day is changing continually throughout the year. Wow. And sometimes it's less than 24 hours. Sometimes it's more than 24 hours. So sometimes the sun gets to its highest point in the sky before clock noon. Right. And sometimes it gets as so what we do is we just average that over the whole year and say sun. You are averaged 24 hours, and there you go. By the way, if you ever look at sundials, there's a map on the sundial that corrects for the sun being early and the sun being late. You either add or subtract up to 14 minutes of the day from what the sun time reads in order to get your clock time. Wow. All right. Every sundial has it. So it's a figure eight.

It's called an analemma. It's got a name, an analemma. But I don't want to talk about analemma right now. So what you have on your clock is the average length of a day over the 365 days of the year. All right. So now we define that as 24 hours, each hour, 60 minutes, each minute, 60 seconds. Got it. Okay. Well, what is defining this? Well, it's the rotation of the earth. Right. Okay. Makes sense. Well, all right. Well, how stable is that? even in the perfect average that we're taking. You'll never know, will you? Because you're using Earth to define the time for me. Right, exactly. If Earth is your measure and you are slowing down, you will never know this. There's no way to know it, okay? Now, back when I was growing up, watches, they would boast, this is how old I am, they would say, get this watch, it's so accurate, it's accurate to two minutes a month.

Okay, that was like accurate that is not a good advertising Okay, well bad then it was because you know cheap watches with ten minutes a month. You'd have to reset your watch every day. Why do you think? Heist movies would always say Let's synchronize our clocks right before the highest that's cool They knew that the clocks there were not keeping good time earth was keeping the best time of them all okay, so now You pose the question Maybe Earth is not the best time-keeping device. Let's offload the responsibility of keeping time to something else, like a vibrating atom. So we did that. That sounds good. The CCM137, there's an electron transition between two energy levels that has a very precise frequency.

Very precise you can measure what's good about that is any lab can get some for CCM and measure this and then define the length of a second in

their lab Okay, okay when you do this You multiply it by 60 you multiply that by 60 you get the hour you multiply that by 24 you do you do this and you find out That earth is slowing down. Well, you know, it's kind of old I'm just saying It's been added. It's tired. It's a long time. It's tired. If you're 4.6 billion years old, you might slow down a little bit, too. You might be tired, too. I'm just saying. You might lose your stuff when you're getting there. That's very sweet of you, Chuck. That's very, very geriatrically sensitive of you. So you only would know that Earth was slowing down once you offload it to something else that tracks time. Right. And what we found out is that the sloshing of tides on the ocean floor on the beach fronts actually works to slow down the rotation of the earth. Oh wow. Okay, because the moon is causing these tides. Right. Okay. And we're rotating faster than the moon is orbiting us. So the moon is tugging on tides backwards on our attempt to rotate. Right. Okay.

So it's almost like there's a counterweight. A counterbalance, exactly. A counterbalance that's pulling against what? Against us. Wow. Tides are slowing down the rotation of the earth. And in response, the moon is spiraling away from us by a couple inches a year. Okay? That's in response to earth slowing down. It all relates to what's called the conservation of angular momentum, but it's a big ballet. Okay, and so we've been slowing down and Chuck that's so we what we could do is say let's redefine the second the length of the second so that we always have 24 hours and 60 minutes and 60 seconds, but that's messy Every year here's a new definition of the second boat folks go to your lab redefine the CC madam. It's much easier to accept the fact that you know, we're breaking up with the moon, but, you know, it's just taking some time. Breaking up is hard to do. Everybody has to be on the same page so that the breakup can be as amicable as possible. So that's a nice way to think about it. So the way we compensate for this is we add a leap second. Oh, well, it has to be every four years, then we add a... No, no, no, no, no, no. Did I mention years?

At all? I'm saying days, dude. Days. Every four days? No, no, no, no, no. What I'm saying is we monitor when we have fell behind my second. Oh, I see. OK. OK. So when the second, when we know that we've lost the second based on our atomic measurement. Correct. Then internationally, we say time to throw in a leap second. Now we throw it in. And by convention,

we throw it in on June 30th or on December 31st. and we can throw in one at each time if we needed it, okay? Since 1973 or four, something like early 70s, there's been 23, 24 leap seconds added to the calendar. Sweet. So when they're gonna add a leap second, they choose which of those days it will be. When they do so, that final minute has 61 seconds in it. Nice! Yes, it's very cool. That's very cool. And so this is what we do. The longest minute of...

Yes, it's the longest minute because it's actually 61 seconds, okay So newspapers like having fun with this a this year will be slightly longer than other years right because you have to throw in the leap second that leap second happens on a Greenwich time so for us it would take place at like I guess 7 p.m. Right because we're five hours Behind them on the green mean time right grace me time right so I said at dinner party once in the June 30th leap second and seven p.m. That's when you're having dinner. And I said pause. I got out the atomic clock and we just watched that sucker tick. Sixty one seconds went by. We toasted it with a sip of champagne and then went on our way. OK. Why are you walking around with an atomic clock? No, no, I have weight. Just no, I have access. Don't just let's not just I got people. OK, right on. All right. So a couple more things.

Other things that can change the rotation of the earth not just the title sloshing, right? Okay, but Here's one for you. You ready? All right. You've seen skaters who want to speed up their speed So what do they do this hand start out extended? And they start with a slight rotation. They bring in their hands and they spin faster, right? What they've done is they change where the mass is relative to their rotation axis. So the farther away the mass is, the slower they're going to rotate. The closer it is to the rotation, the faster they'll turn. If Earth has an earthquake and one of the continental slabs shifts north, then there's mass on Earth's surface that used to be closer to the equator the day before and is now moved closer to the pole.

that will have the effect of speeding up the rotation of the earth. Large animals that migrate will change where the mass of the earth is from the northern climes down to the south and they're farther away from the rotation of the axis. So not only does do tides affect the rotation of the earth, seasonal migration of animals affects the rotation of the earth, and so does earthquakes. so do volcanoes and anything else that's remapping the surface

of the earth. And you know what else to change it? The melting of land glaciers. Uh-oh, we're in trouble. Okay. You got Antarctica has land glaciers as it melts. What used to be right there near the pole now melts and goes closer to the equator. That will slow down the rotation of the earth because that's like the skater bringing their hands farther away.

All right. So all of these factors combined, basically Earth's day has been increasing rather than de-imprinciably could decrease, depending on what's going on. Then I had this diabolical idea. We'd go set up jet engines, anchor them to the Earth, face them either due east or due west, and ignite them, and then the exhaust would help speed up or slow down the Earth. I thought we might do that. Then I did the math on that, and no, it's not going to work. It's not enough. You would have made a good bond, villain, though. Yeah, it's just, you know, it's like Lex Luthor saying, let me create an earthquake to make these front property in California. So yeah, it's geoengineering on a huge scale. But in principle, you could do that.

But if you take the greatest engines and fire them, the mass of the Earth is so huge. You just have essentially no effect. No effect at all. Yeah. So I'm just saying the rotation rate of the Earth is susceptible to all these small effects that can add up. And we deal with it. We deal with it. We throw in leap seconds. 23 hours, 56 minutes and four seconds. That's our rotation rate. to the stars. They start. It's exactly 24 hours on average, and that average is changing. Right. Oh, man, that's so cool, man. Yeah, just to show you that smart people think about this stuff and solve it so you don't have to, because you just think everything, oh, just it works. Oh, the computer tells me what the right time is, and everything in the bank knows what time it is, and my GPS, we've got top people thinking about this stuff. So before the time of clock, Nobody knew what time it was. That's why they synchronized clocks at the beginning of Heist movies. Everyone had different time, dude. Oh my God, that's crazy. Oh man, I wish I lived back then. I'd never be late for anything. It'd be like Chuck, you're 10 minutes late. Not according to my watch. And as far as I could tell.

Don't nobody know what's happening. I'm on 10 minutes late. You 10 minutes early. That's what the problem is. So that's all I got to say. That is a fascinating thing. That is fascinating. And I love thinking about it just

because of how many different branches of the science and technological world had to come together to figure this out. Sweet. Well, I got to go. I'm preparing for the next leap second. You're gonna be like... So, all right, dude. By the way, if we have a leap second announce, we'll do a special show and maybe celebrate that and I'll bring out my atomic clock and we can watch it. That'd be so cool. I'm all about it. Okay. Yes, and make sure I don't forget the champagne because you said you had champagne the last time. So don't even try to skip out on the champagne. You forget that? That's right. Okay, I'm just saying. Hi, I'm Ernie Carducci from Columbus, Ohio.

I'm here with my son, Ernie, because we listen to StarTalk every night and support StarTalk on Patreon. This is StarTalk with Neil deGrasse Tyson. Consider a few things. I want to talk about life and, yeah, it's relevant, and on size and life. Well, listen, the way I see it is this, okay? You know, size is a preference thing, okay? And not all of us can actually, you know, be what some people might have as a preference. And, you know, why are you looking at me like that? Because why? Perhaps you've been told something about, you know, my ethnicity and maybe I'm supposed to be a certain way. Well, it doesn't necessarily happen that way for every person in my position.

someone who looks like me. We are witness to Chuck digging himself out of a box. So here we go. So you're in a room there, right? All right. Looks like a hotel room. All right. You can walk around on the floor. OK. If there were an aunt walking around on the floor, it would be doing just what you're doing. Absolutely. However, an aunt can go to the wall and then just walk up the wall. Right. How come you're not walking up the wall? Because I'm not Spider-Man. Okay. That's all. If I were Spider-Man, I'd be walking up the wall. Okay. Because Spider-Man is imitating a spider. Yes, he is. Okay. But there are no spiders the size of humans. This is true. That can just walk up the wall. Correct. All right. There's a reason for that. There's a very good reason for it. And it has to do with the physics of life. Nobody ever talks about it that way. People say, oh, is the biology this? You can talk.

Any biology you want at the end of the day, physics sets the rules. Take that biologist. His little smackdown for you, biologist. Physics sets the rules, even in your world. Physics. All right. So it all comes down to this. Go ahead, Jack. OK. So how strong are your limbs? Well, the strength of your limbs is measured by the cross-sectional area. Oh, God. So that tells you how big your muscles are. Well, you're just embarrassing me on every level today. First, we're talking about size. Now, we're talking about skinny wrists. Thanks. All right. So just take any part of your body, your legs and your arms. Take the cross-section. And that tells you how big your muscles are. And when you say big muscles, you're measuring sort of circumference, OK? And that's related to the circumference of the area. So as you get bigger, your strength will grow as the area of your muscles, the cross-sectional area of your muscles. OK. However, your weight goes up as the cube of those dimensions. Oh my goodness. Now, of course, I knew your weight goes up, but I didn't realize it. Wow, that's a lot. That's OK. So that's why a 300-pound person does not look all that much bigger than a 200-pound person.

Right. Okay, the chubbier and they're big, you know, if they could be tall, whatever. But if you start increasing depth, height and width, right, your volume, as your volume increases, so does your mass, but your strength necessary to sustain that volume and that weight does not go up in proportion. Right. So your limbs, if you're going to hold up a heavy animal, Your limbs have to be much fatter than they would otherwise be if you were smaller. And what are your heaviest animals on earth? Your elephant, your whale. Elephant. Yeah, on Earth's surface. The way, the elephant. The elephant. Its legs are huge. Yes. Okay. They don't have spindly legs. No, they don't. Their legs are huge to compensate for the fact that strength only goes up as the area of their muscles while the weight is going up as the volume and volume rapidly outstrips area. Okay. Now you can go the opposite direction as you get smaller and smaller and smaller. Your the weight with your volume gets lower and lower and lower and the strength of your limbs becomes better relative to your body weight. So I can lift. What is it? Not only is it why they can lift, things relative to their body weight. Relative to the body weight. Okay. All right. Can they do that? It's why they can get around with really skinny legs. Right.

Okay. Yes. They have six of them, but they're really skinny compared to the size of the thickness of their body because the legs aren't holding up much weight relative to their own strength. Right. And they intermittent fast. They do intermittent fasting too. That's how that works. Thank you. Go to an egg colony to have these diet books. Wow. So, you know what? You just, you just gave me a really stark thought. which is I watched this documentary about this guy who climbed the only cliff face that these people die all the time climbing it without any bailay. And this guy was super skinny. Yep. Yep. Not. Not strong like with big giant muscles, but I mean his muscles when you look at his musculature This dude was ripped. He's ripped, but he doesn't weigh very much. He doesn't weigh much at all. Correct. Correct. And that's why in the movie Cliffhanger, which involve rock climbers and mountain climbers, it starred Sylvester Stallone, who's got all these muscles. It's like, no, every mountain climber on earth knew that that is not the body type.

because he's got muscles where he does not need them. He's got body mass that is counter to his ability to support himself. So the point is generally, and there's certain exceptions depending on environments, the smaller you are, the less you weigh, the thinner your legs can be and still hold up your weight. And the bigger you are in mass, the thicker your legs are. So rhinoceroses and hippopotamuses, hippopotamide and elephants all have really thick legs because as they got bigger, if they kept their thin legs, they break their legs. And that's the end of you. You get eaten and you leave the gene pool very quickly when that happens. All the lions are missing you. Remember what we used to...

Remember those elephants with the skinny legs, man? We used to just find them lying around. Oh, they made such a delicious meal. Remember those days? We would just walk out on the plane and we would find the elephants. The elephants are crutches because they kept breaking their legs. Right. So anyhow, so snap. So that's why you don't find very tiny creatures with really thick legs. That's that. So because they don't need it, that's the point. OK. So so now that so that's an important fact. Number one, whales are different because they don't weigh anything. Right. You got to go back to our weight density and whatever I was just about to say we talk about I think we talked about no we didn't but they're buoyant because they're at a

certain depth and that right So they don't they so they don't have to hold up their own weight. That's why they can get so big right Okay, so it doesn't it's not fair to say how much is that well it weighs 400 tons No, it doesn't it weighs zero because it is neutrally buoyant in the water. And if you bring it onto the land, it will die because it cannot hold up its own weight. It cannot do anything in its own weight. So you cannot say it weighs that much because that's not where it lives. It makes so much sense because you've never seen a beach whale say, all right, I'm out of here.

I'm good with the land thing. I tried the beach. I tried the beach. I'm gonna roll myself back on it. Yeah, I'm good. You never saw it. You never saw it. So they don't weigh anything. They don't have to hold up their own weight. So they're really not. So everything I'm describing is sort of in the air, on Earth's surface. Okay? All right. So, but I'm not done. Oh, cool. Not only that. The laws of physics don't all manifest equally at all size scales Interesting, okay, so If you're gonna drink water you'll put it in a glass and then you'll drink it like this, right? Why do you put it in a glass? Why don't you just sort of pour it in front of you? Well, because it'll flatten out on the table and you'll make a mess, right? Okay, but suppose I am an insect again, let's suppose I'm an ant Does an aunt need a vessel in which to put the water in order for it to drink? No, because water beads up. There's a bead of water that does not continue to spread. It could just walk up and suck the juice right out of the bead of water until it's gone. Okay. Now, the producers and writers of the movie Bugs Life knew this. So when the mosquito went to the bar and ordered a drink, And what drink did the mosquito order? A Bloody Mary, of course! Okay, so Bloody Mary, O-positive. I just turned it to my mother, I can't believe it. You turned it to your parents, that's right. I turned it to my parents then. Oh my God. Okay, so Bloody Mary, O-positive, okay? And so the bartender...

which is another insect, I don't remember what kind, has a spigot, pours out a blob of Bloody Mary and plunks it down in front of the mosquito and it's just a ball, okay? Held up by surface tension. Surface tension is this sort of, this imaginary film that exists on the surfaces of liquids that you have to sort of puncture that in order to fall through. But that sort of film actually contains it in small enough amounts. Service attention is not strong enough to contain a big blob of water like this because gravity wins. But when

you're small enough, gravity loses against service attention. Service attention wins. They plop this drop of Bloody Mary in front of the mosquito, mosquito put in its nose, sucked it out, and then that was the end of that scene. This was brilliant. And it's in that movie. It's a Pixar movie that people who knew physics and chemistry right in the script for that movie. So different forces operate when you're small than when you are large. And so that's why they have, there's something called, is it a Jesus spider, a water strider? There's one something that walks on the water swiftly. And in some regions, they call it a Jesus spider, I think. Anyway, because Jesus in the Bible walked on water, except...

Jesus weighs more than the surface tension would have held him up. So Jesus would have just sunk without some kind of magic or spirit or miracle. Whereas if you're small enough. Well, maybe he had just giant foam sandals. Foam sandals. We didn't check. He was the first person to wear the giant foam sandals. And they're just like, oh my God, Jesus. It's the miracle. People misinterpreted it. Right. They were like, hey, go ahead. So all I'm saying is that if you're small enough, that surface tension, you don't weigh enough to break the surface tension. So then you can just sort of walk on the water, as some insects do. So my point is, size is everything in this world. And you might say, well, could you have something really big, like life form the size of a galaxy?

Probably not. Here's why. Because suppose you're that big, by the way, the galaxy is 100,000 light years across. Suppose you have your head itches, and then you send the signal to the brain, and then it goes to your finger, and you got to bring your finger up to scratch your head. That took about 30 million years. It's going to take at least 100,000 years to get your arm to your head moving at the speed of light. And you're probably not moving at the speed of light. So you're not going to be able to respond to stimulus the way life needs to in order to thrive in an environment. So there's certain sizes above which are not realistic. So physics contains the largest and the small. And by the way, you can't be so small.

That quantum physics prevents your molecules from from binding you can't be smaller than the sizes of molecules, right? Because what are you at that point? Okay, if molecules comprise life You can't be like that smaller than a

molecule. It's just not there's no so so the chemistry and physics contain all the realms in which you will find Biology and that's just the beginning of it. I'm giving you the basics of it where that's the entry level sort of size and life 101 So so there you have it. So so so so spider-man he's got the powers of a spider No, not if he's that size. No doesn't know he doesn't know he doesn't all right. No, he doesn't He can't No. He basically just has the powers of a really good rock climber. Rock climber? Yes. That's about it. That's about it. All right. We got to go there. Good to have you. Oh, it's a pleasure. Neil deGrasse Tyson, signing off, bidding you. Keep looking out.